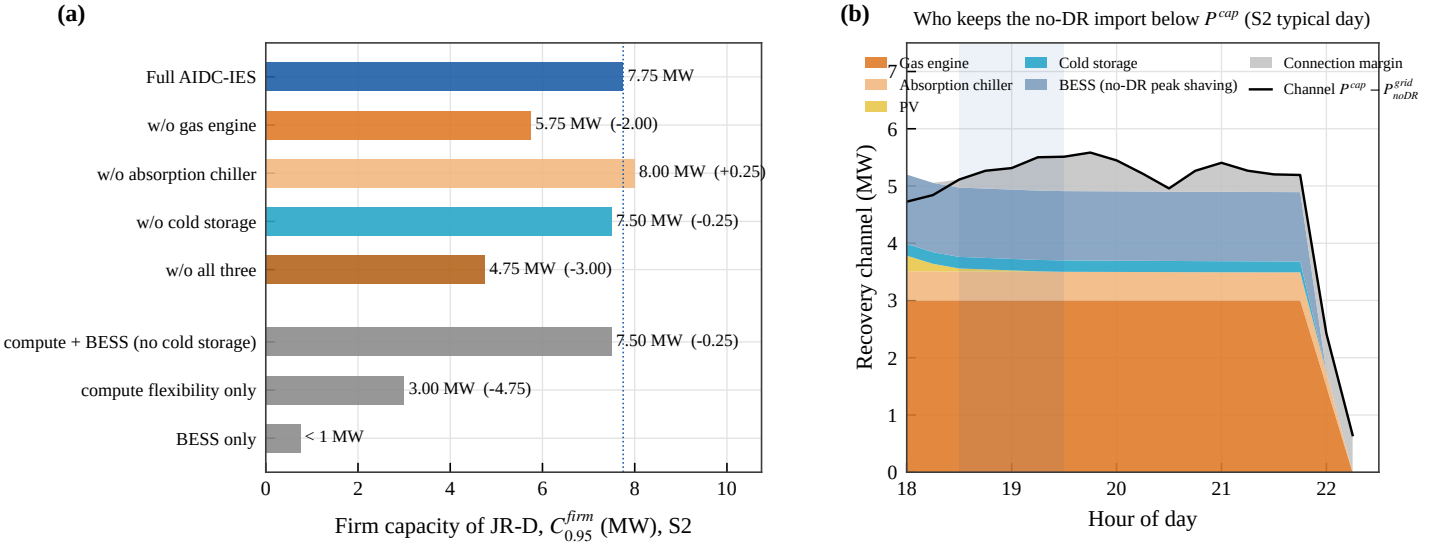
(a)
Full AIDC-IES
w/o gas engine
w/o absorption chiller
w/o cold storage
w/o all three
compute + BESS (no cold storage)
compute flexibility only
BESS only
7.75 MW
5.75 MW (-2.00)
8.00 MW (+0.25)
7.50 MW (-0.25)
4.75 MW (-3.00)
7.50 MW (-0.25)
3.00 MW (-4.75)
< 1 MW
0
2
4
6
8
10
Firm capacity of JR-D, $C_{0.95}^{firm}$ (MW), S2
(b)
Who keeps the no-DR import below P^{cap} (S2 typical day)
Gas engine
Absorption chiller
PV
Cold storage
BESS (no-DR peak shaving)
Connection margin
Channel P^{cap}
P_{noDR}^{grid}
Recovery channel (MW)
0
1
2
3
4
5
6
7
18
19
20
21
22
Hour of day